

Embedding Models

Wednesday, July 29, 2026 7:55 PM

Model	Dimensions	Advantages	Use Case
OpenAI <code>text-embedding-3-small</code>	1536	Good quality, lower cost	General enterprise search
OpenAI <code>text-embedding-3-large</code>	3072	Highest quality	High-accuracy RAG
BAAI <code>bge-large-en-v1.5</code>	1024	Strong open-source model	Self-hosted deployments
<code>bge-m3</code>	1024	Multilingual + dense/sparse support	Global enterprise systems
<code>e5-large-v2</code>	1024	Excellent retrieval quality	Technical documentation
<code>all-mpnet-base-v2</code>	768	Lightweight	Smaller deployments

Recommendation

Cloud Solution: OpenAI text-embedding-3-small

Why? High quality, Lower cost, Fast inference, Suitable for millions of chunks

Self-hosted Solution: BGE-M3

Why? Open source, Multilingual, Excellent retrieval performance, Supports hybrid retrieval (dense + sparse)

Popular embedding models:

- OpenAI text-embedding models
- BAAI BGE
- Nomic Embed
- Jina Embeddings
- Voyage AI
- Cohere Embed

The main difference is **how documents and queries are represented** and **what kind of similarity they capture**.

Feature	Dense Vector Search	Sparse Vector Search
Representation	Dense embeddings (hundreds/thousands of floating-point numbers)	Sparse vectors (mostly zeros, only a few non-zero dimensions)
Generated by	Neural embedding models	TF-IDF, BM25, SPLADE
Captures	Semantic meaning	Exact keywords
Handles synonyms	✅ Yes	❌ Usually no
Handles paraphrases	✅ Yes	❌ Weak
Explainability	Lower	Higher
Speed	Good with ANN indexes	Very fast with inverted indexes

When to Use Each

Use Dense Search when:

- Building RAG systems
- Question answering
- Semantic document search
- FAQ retrieval
- Natural language queries

Use Sparse Search when:

- Searching product IDs or SKUs
- Log search
- Error codes
- Legal documents with exact terminology
- Code search
- Exact keyword matching is critical

8919619373 - katari seethal samson
9811876495 - Ajay kumar
9004194946
9137503867 - shetal kaniyalal mehta
9150806968 - ramki (unionys) spoked
7358328818 - deepika
9307043320 - hexaware spoked